# Dengue graphs of 2008 to 2016

By Troy and Intil

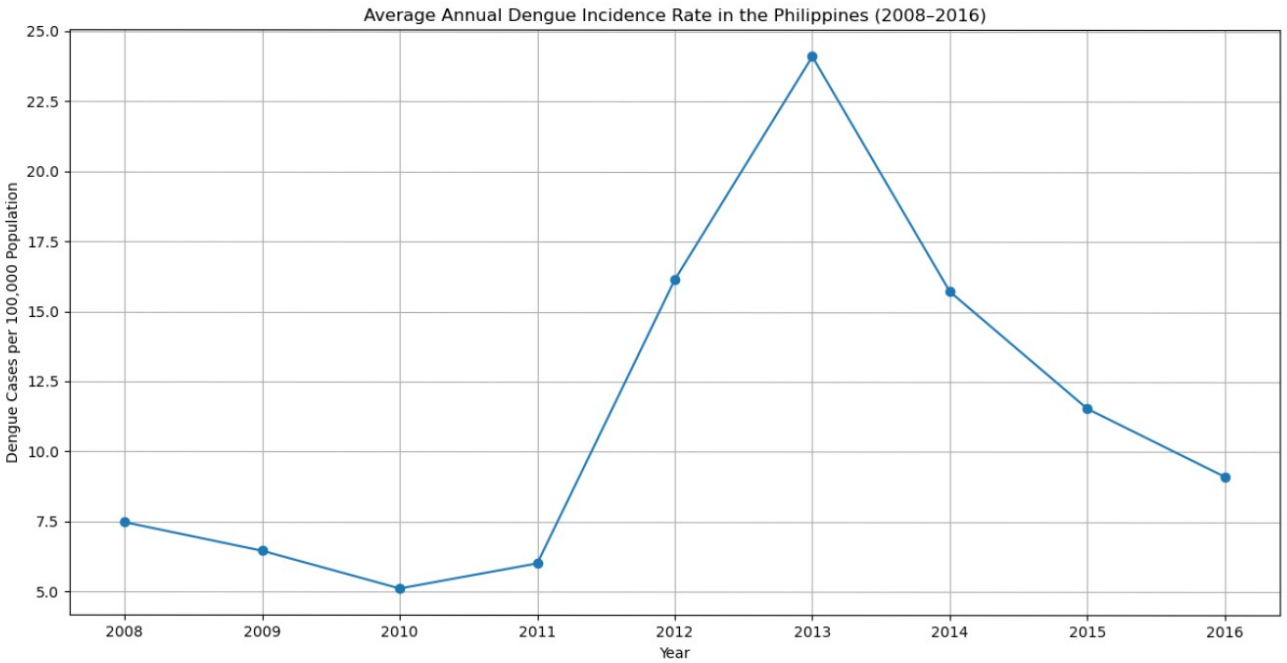
# The data

- This Dataset contains the recorded number of dengue cases per 100,000 population per region of the Philippines from 2008 to 2016. We used python with matplotlib and pandas module to read, clean, and visualize our Dataset.

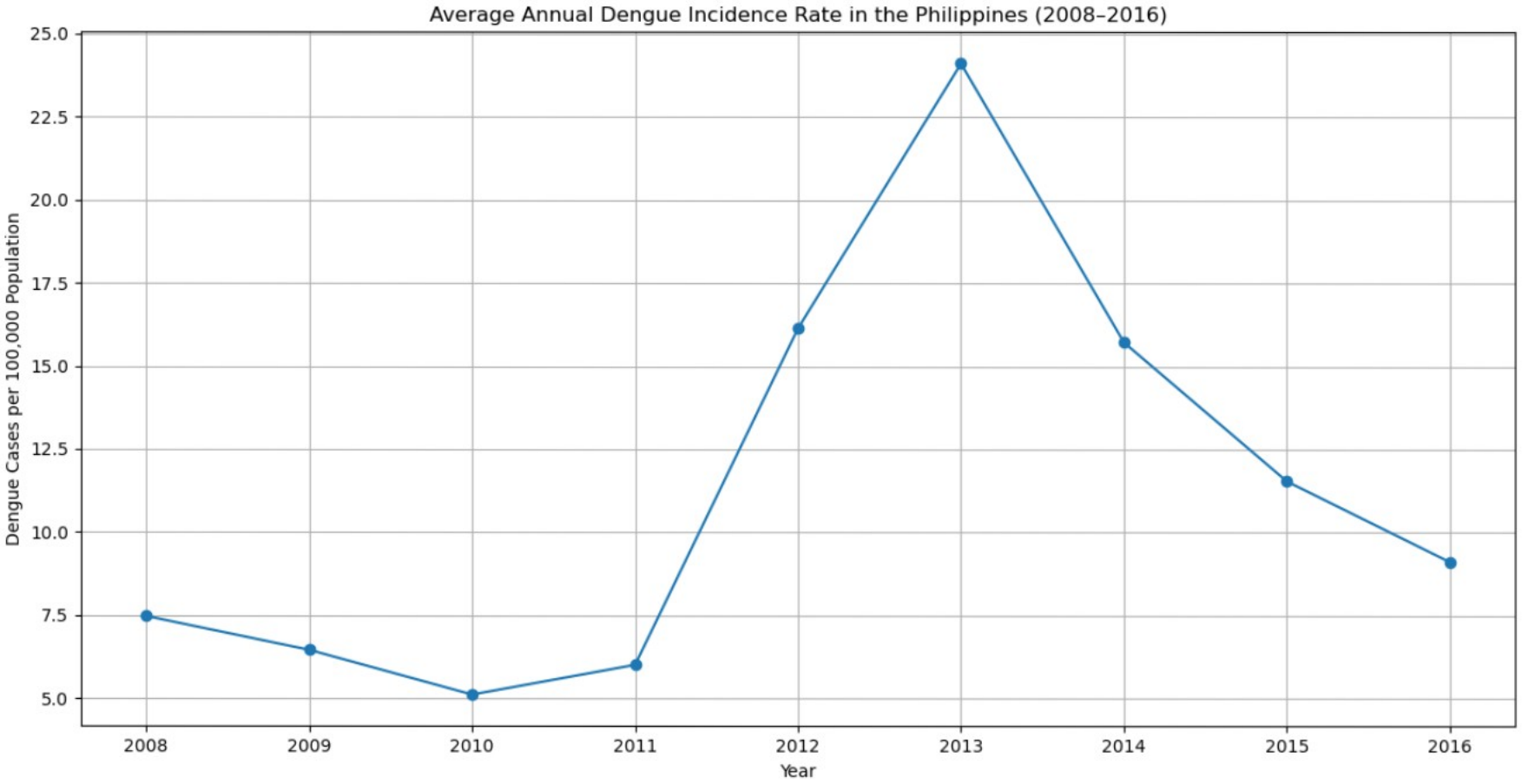
| Month | Year | Region | Dengue_Cases |
|---|---|---|---|
| Jan | 2008 | Region.I | 2.953926 |
| Feb | 2008 | Region.I | 2.183336 |
| Mar | 2008 | Region.I | 0.97241 |
| Apr | 2008 | Region.I | 9.357156 |
| May | 2008 | Region.I | 7.320599 |
| Jun | 2008 | Region.I | 4.513452 |
| Jul | 2008 | Region.I | 6.82522 |
| Aug | 2008 | Region.I | 9.210377 |
| Sep | 2008 | Region.I | 13.301839 |
| Oct | 2008 | Region.I | 1.724652 |
| Nov | 2008 | Region.I | 1.32101 |
| Dec | 2008 | Region.I | 1.0458 |
| Jan | 2009 | Region.I | 5.23598 |
| Feb | 2009 | Region.I | 3.719351 |
| Mar | 2009 | Region.I | 2.473549 |
| Apr | 2009 | Region.I | 3.051313 |
| May | 2009 | Region.I | 4.658217 |
| Jun | 2009 | Region.I | 6.644278 |
| Jul | 2009 | Region.I | 1.263857 |

# Line Graph.

- This shows the average annual dengue incidence rate per 100,000 population across all Philippine regions from 2008 to 2016. The trend reflects changes in dengue intensity over time, independent of regional population size.

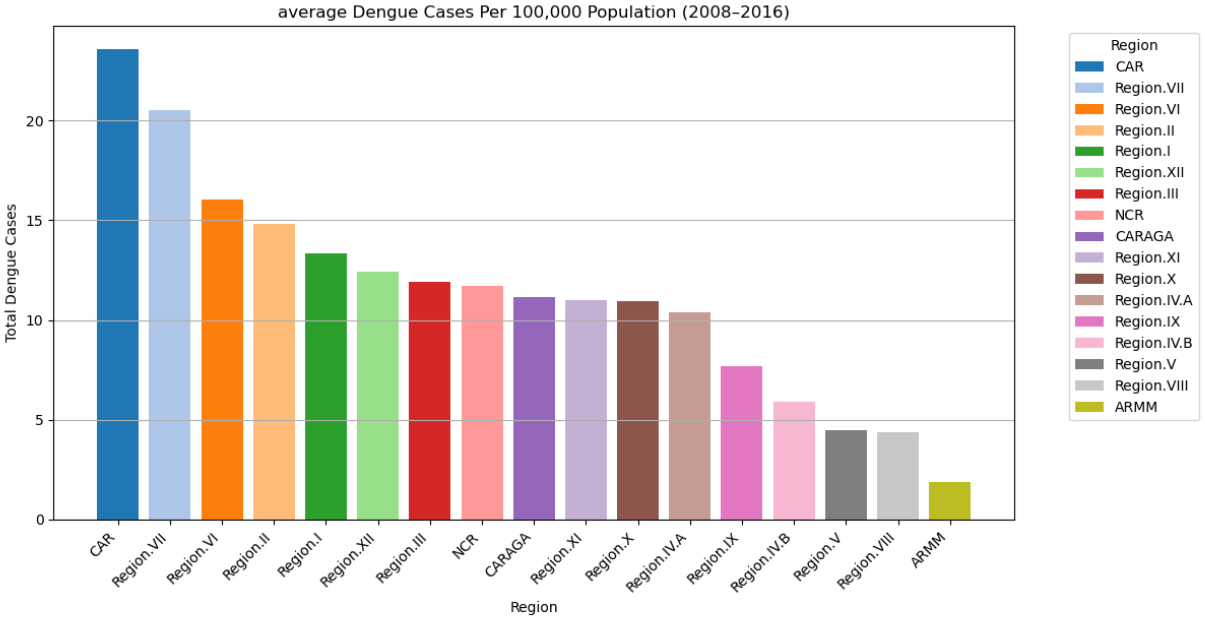


# Line Graph.


Average Annual Dengue Incidence Rate in the Philippines (2008–2016)
Dengue Cases per 100,000 Population
25.0
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2008
2009
2010
2011
2012
2013
2014
2015
2016
Year

# Per region bar Graph.

- This shows the cumulative dengue incidence rate per 100,000 population for each Philippine region from 2008 to 2016. Higher values indicate regions that consistently experienced higher dengue incidence relative to their population size.



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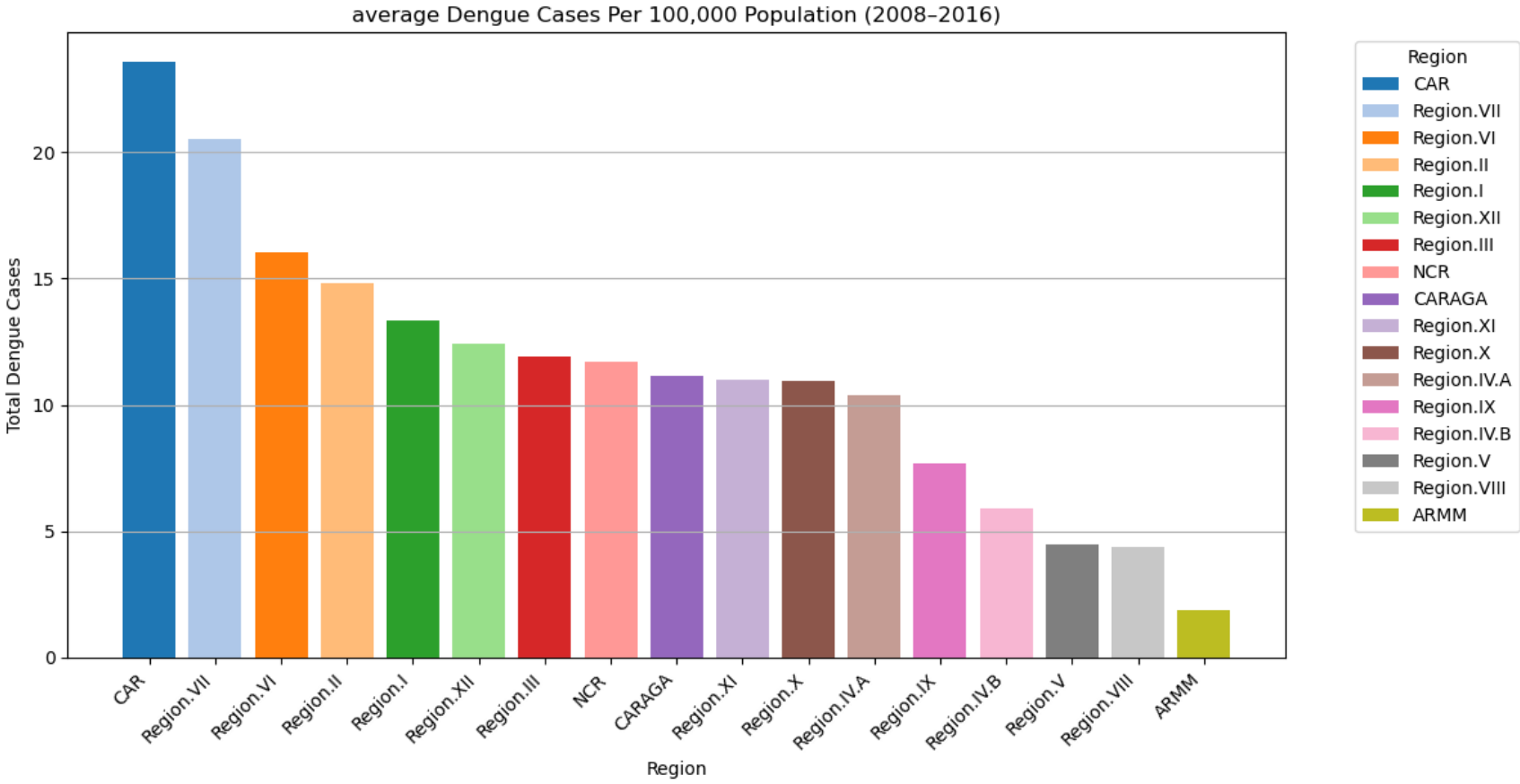
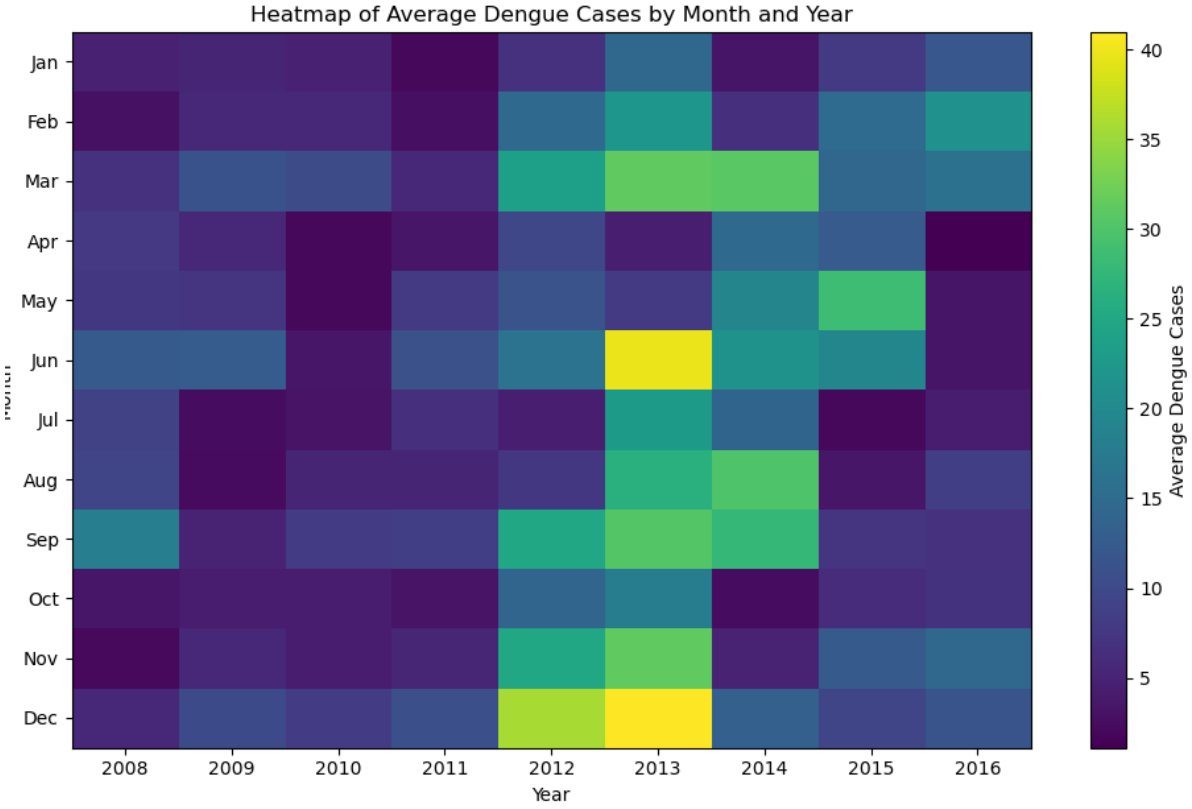
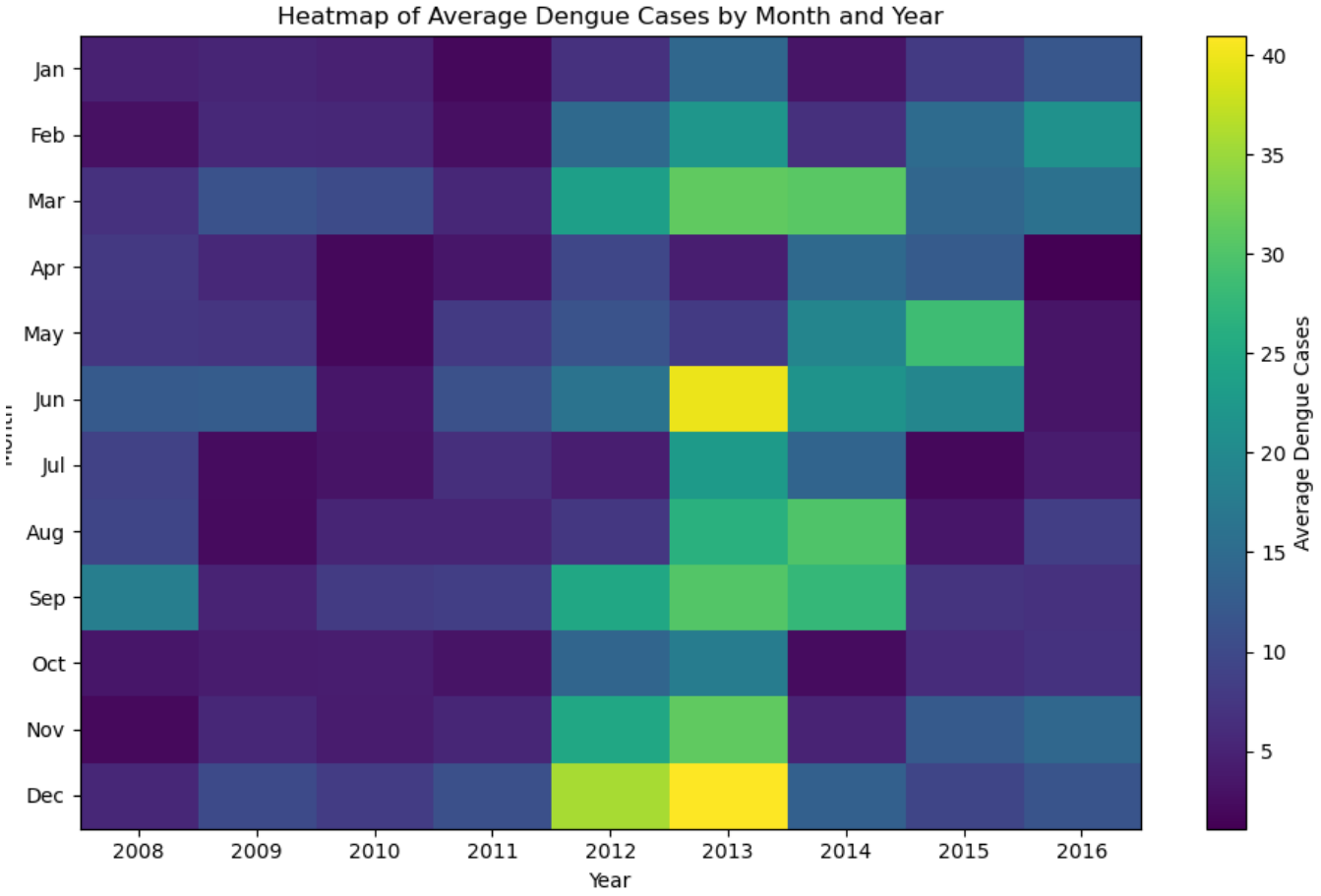


# Heatmap

- This heatmap shows the average number of dengue cases by month and year. Each cell represents how severe dengue cases were for a specific month in a specific year.



# Heatmap


Heatmap of Average Dengue Cases by Month and Year
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
2008
2009
2010
2011
2012
2013
2014
2015
2016
Year
40
35
30
25
20
15
10
5
Average Dengue Cases

# Why Line Graph?

- Show changes over time
- Clearly display year-to-year trends in dengue incidence
- Make it easy to identify peaks, declines, and long-term patterns
- Supports comparison across years without visual clutter

# Why Bar Graph?

- Compare regions
- Bar graphs allow direct comparison between categories
- Effectively highlight regional differences in dengue burden
- Summarizing data as averages reduces noise from monthly fluctuations
- Easy to interpret for both technical and non-technical audiences

# Why Heatmap?

- It handles Two Variables of Time simultaneously
- . Spotting "Seasonal Drift"
- Identifying "hot spots" & seasons.

# Why not other Charts?

- Pie Charts are not good good because there are too many years and many regions.
- A standard scatter plot usually compares two continuous variables (like height vs. Weight, X axis and Y axis).
- Radar charts become unreadable "spaghetti" once you add more than 3 or 4 variables

# Goodbyezzzzz

- Source Codes for learning opps:
  - https://github.com/AlieeLinux/Visualization-Course-IDK